# Solution overview

An analyst workspace for portfolio monitoring and explainable investigation.

## ANALYST WORKFLOW

**Portfolio:** filter region, product and employment; inspect income/age cuts, observed exposure, cohorts and employment x income. **Account:** inspect retrospective score, signed drivers, repayment history, undated customer bureau lines and collections. **Ask:** query exact aggregates and inspect evidence. **Review:** investigate flags and record confirm/dismiss decisions in SQLite and export CSV.

## MEASURED MODEL RESULTS / HELD-OUT TEST

**Selected logistic regression:** AUC **0.6768** | KS **0.2607** | Gini **0.3536**. Histogram boosting: AUC 0.6613, KS 0.2416, Gini 0.3226. Test: 2,948 loans / 472 defaults. Customers split 60/20/20; train-only imputation/encoding; validation AUC selects the model. Bureau + repayment features; global permutation importance and signed logistic drivers. IDs, gender and collections excluded.

**Interpretation:** retrospective classification only, not a validated future PD. The target has no default date or horizon; bureau snapshots lack reporting dates. Monitor input/score drift and delayed-label AUC/KS/Gini/calibration after a prospective design is established.

## TOP THREE PORTFOLIO INSIGHTS / FULL SYNTHETIC SAMPLE

**Mortgage:** observed default rate **24.65%** (353/1,432 accounts). Review underwriting and monitoring for this product; observed association is not causation.

**Unemployed:** observed default rate **34.96%** (251/718 accounts). Investigate affordability and repayment patterns before changing credit policy.

**Unemployed / <25k:** observed default rate **44.62%** (87/195 accounts). Investigate the joint affordability/employment segment; aggregate employment averages may hide this concentration.

## TWO EMBEDDED AI FEATURES / HUMAN JUDGMENT REQUIRED

**Ask the Portfolio - trigger:** analyst submits a question. Groq proposes a constrained query; Python computes numbers and provenance. Unsupported questions are declined. **False positive:** wrong interpretation yields a valid query. **False negative:** supported query is refused. **Human step:** verify interpretation, filters and exact source figures. Only definitions, categorical scope and question reach the provider.

**Anomalies - trigger:** each build runs peer rules and Isolation Forest; current queue has 1,685 accounts. Flags show payment, DPD, missed-payment or bureau evidence. **False positive:** unusual legitimate behavior. **False negative:** missed deterioration or risky peers masking a problem. **Human step:** inspect history, confirm/dismiss with a note; SQLite keeps the versioned decision; CSV export retains a copy. Unusualness is not default probability.

## ASSUMPTIONS & PRODUCTION NEEDS

Latest observed outstanding is exposure, not a common-date bank balance. Twelve repayment months per loan; synthetic chronology includes collection events before origination, retained as quality warnings. No causal claims. Anonymous public demo reviews need authenticated identity in production. IT/data teams must provide timestamped snapshots, cutoff/outcome definitions, approved vendors and access, monitored hosting, backups, audit controls and model-risk review. SQLite files may be lost on hosted restart/redeploy; export CSV. Managed Postgres is optional for durability. Hosted link, repository and video are separate submission steps.